

A Two-Stage LSTM-LightGBM Framework for Time-Series Traffic State Prediction

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Abstract—Urban mobility plays a significant role as the backbone of social and economic growth. However, the high urban mobility that corresponds with high population growth increases vehicle volume. Consequently, the road capacity cannot accommodate the overloaded vehicle volume, resulting in traffic congestion at various levels. Therefore, as an effort to improve human commuting life, this paper proposes a two-stage LSTM (Long Short-Term Memory) and LightGBM (Light Gradient Boosting Machine) framework for binary-class traffic forecasting based on time series data. The proposed algorithm constructs an LSTM model to capture the temporal dependencies embedded in sequence vehicle count data, and a LightGBM algorithm is employed for classifying the traffic state into two binary classes: Dense Traffic and Normal Traffic. The evaluation results demonstrate that the proposed algorithm achieves an RMSE and MAE of less than 0.2, indicating a time series prediction with minimal error. From a classification perspective, the proposed algorithm achieves an accuracy of 83.4%, outperforming other machine learning algorithms.

Keywords—LSTM, LightGBM, Traffic Prediction, Binary Class, Time Series Prediction.

I. INTRODUCTION

Currently, the demand for citizens' mobility in fulfilling basic life needs is increasing dramatically. In this case, the high and rapid mobility contributes to traffic congestion, leading to speed reductions and increased travel times [1], [2], [3]. The situation is worse in fast-growing cities, such as Jakarta, where the average travel speed drops to its lowest levels of 15 km/h during peak periods [1]. Several factors affecting this condition include the rush hours, collisions, maintenance work, and unfavorable weather [4].

Poor traffic conditions also affect businesses' operations, particularly those that heavily depend on transport logistics [5], leading to increased fuel consumption and travel expenses [6]. Moreover, congested traffic increases global carbon dioxide (CO₂) emissions by 25% to 30% [1]. Accordingly, an

intelligent data-oriented solution is necessary to handle traffic congestion [3], [7].

Over the past decade, numerous studies have been introduced to tackle traffic prediction. Kafy et al. [8] investigate the use of conventional machine learning algorithms for predicting traffic congestion based on historical vehicle count data. This research utilizes an open-source dataset containing traffic records for various junctions and compares five supervised machine learning algorithms: Decision Tree, Random Forest, Logistic Regression, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). Each model is then evaluated using the Mean Absolute Error (MAE) metric to establish its prediction accuracy. The evaluation shows that logistic regression achieves the lowest MAE of 15.38, outperforming other models. This study highlights the utility of lightweight models for time-series prediction, particularly in situations where computational efficiency is crucial. However, this method is limited by its reliance on traditional models, which are not designed to capture complex temporal dependencies or provide class outputs for categorizing traffic conditions.

In another study, Bao et al. [9] address the issue of traffic forecasting under different weather conditions by enhancing a deep-learning model with weather data inputs. A Deep Belief Network (DBN), composed of stacked Restricted Boltzmann Machines (RBMs), is used for hierarchical feature extraction, followed by Support Vector Regression (SVR) to perform the final prediction. While the hybrid DBN-SVR model outperforms baseline models such as ARIMA and a perceptron-based neural network, its architecture remains focused on numerical prediction. It does not provide direct classification into actionable traffic categories.

Lu et al. [10] proposed a hybrid short-term traffic flow prediction model that combines ARIMA for linear trend extraction and LSTM for learning nonlinear temporal relationships. Their model outperforms several baselines in terms of MAE, MSE, RMSE, and MAPE across multiple

highway datasets. However, this approach also targets the estimation of continuous vehicle flow. It lacks a dedicated classification layer to translate results into clearly defined traffic states, which are critical for real-time traffic control and decision-making.

A hybrid CNN-LSTM algorithm has been introduced to facilitate real-time prediction [11]. The model operates within an edge computing environment that gathers inputs from connected vehicles and smart devices, enabling rapid on-site analysis. By combining spatial and temporal modeling, the system is structured to support real-time traffic prediction for innovative city applications. However, CNN-LSTM models typically merge feature extraction and classification into a single process, which limits their ability to differentiate between closely related traffic states, such as gradual congestion and abrupt incidents.

In addition, a heterogeneous retentive network has been proposed for short-term traffic speed prediction [12], where multisource urban data are combined into high-dimensional features that preserve geospatial relationships. A GRU module further models temporal dependencies, while inputs such as social media, real estate, and points of interest provide richer context. However, this direct mapping from heterogeneous features to predictions can introduce noise sensitivity and reduce generalization when specific data sources are sparse or inconsistent.

Recently, a graph neural network has been proposed to improve traffic prediction by capturing road network information, modeling long-term dependencies, and handling uneven data distribution [13]. Multi-layer synchronous graph convolution learns local spatiotemporal features, while temporal self-attention and trend-aware multi-head attention capture broader dynamics. Nonetheless, the tight coupling of sequence learning and classification in GNN-based methods restricts their flexibility to refine learned representations for accurate traffic state discrimination.

To address the limitations identified in previous studies, this paper presents a two-stage hybrid model that integrates Long Short-Term Memory (LSTM) networks for temporal sequence modeling with LightGBM for final classification of traffic states. Unlike conventional LSTM-only or LightGBM-only approaches, which directly map temporal features to output states, the proposed algorithm separates sequence representation learning from classification. The LSTM algorithm focuses exclusively on capturing temporal dependencies in evolving traffic flows, specifically peak-off-peak transitions and congestion buildup. LightGBM subsequently improves these representations to classify traffic states more accurately. This division provides two traffic-specific advantages: (i) LightGBM improves discrimination between similar but distinct traffic patterns (e.g., recurrent congestion vs. sudden incidents) by using boosted decision boundaries, and (ii) its tree-based structure can handle heterogeneous inputs standard in traffic data, such as mixed sensor readings, missing values, and abrupt fluctuations, thereby enhancing the traffic state predictions. Accordingly, our main contributions are summarized as follows:

1) *Traffic Time Series Prediction* : This paper proposes a time series prediction based on the last ten sequences of vehicle data. By utilizing the LSTM framework, the proposed algorithm captures short-term temporal dependencies to

forecast future values. Here, four different types of vehicles are predicted: bikes, cars, trucks, and buses.

2) *Binary Class Traffic State Classification* : This paper shifts the paradigm from traffic flow regression to categorical traffic state classification, addressing a critical gap in prior work. Our model outputs labels that are more actionable for real-time urban traffic control systems, i.e., dense traffic flow and normal traffic flow. This reformulation enhances interpretability and utility in innovative transportation applications, enabling authorities to deploy adaptive responses based on discrete traffic conditions rather than raw flow values.

3) *Flexible and Robust Hybrid Pipeline* : The proposed algorithm introduces a hybrid architecture that combines an LSTM for temporal feature extraction with LightGBM for robust classification. Unlike fully end-to-end machine learning models, which can face technical challenges such as overfitting, instability, or reduced interpretability, this pipeline allows each model to focus on its strengths. LSTM captures sequential patterns in the traffic data, while LightGBM classifies the learned features with high accuracy. Evaluation results show that the proposed algorithm outperforms other algorithms.

To the best of the authors' knowledge, no prior study has proposed a two-stage hybrid framework for simultaneously modeling and classifying traffic in sequential prediction tasks. By modeling traffic patterns temporally and translating them into meaningful traffic condition categories, the proposed method facilitates intelligent traffic control, enhances commuter experience, reduces environmental impact, and supports innovative city development. The remainder of the paper is organized as follows. Section II presents the architecture model, Section III presents the proposed algorithm, Section IV presents results and discussion, and the Conclusion is summarized in Section V.

II. ARCHITECTURE MODEL

This study proposes a hybrid model that integrates LSTM with LightGBM to enhance the temporal precision and classification accuracy of traffic states. Fig. 1 shows the flowchart of the proposed algorithms. The proposed architecture integrates sequential time series forecasting with classification to enhance the analysis of traffic data. The pipeline begins with the acquisition of raw traffic data, followed by a crucial data preprocessing stage. Subsequently, an LSTM network is employed to capture temporal dependencies. Once the LSTM model generates the predicted time series data, these outputs are forwarded to a LightGBM classifier. LightGBM then processes the forecasted features to perform classification tasks, i.e., categorizing traffic flow conditions. This proposed hybrid approach utilizes the strengths of both deep learning and tree-based learning.

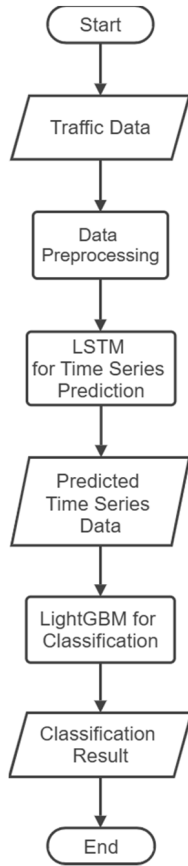


Fig. 1. Flowchart of the proposed algorithm.

III. PROPOSED ALGORITHM

The following sections outline the step-by-step implementation process, including data preprocessing, model parameters, and architecture. This section also covers the details and results of hypertuning parameters

A. Dataset Preprocessing

The study employs the "Traffic Prediction Dataset," an open-source resource available on KaggleHub [14], specifically designed for analyzing and forecasting urban traffic conditions. This dataset contains daily traffic observations over two months, with data collected at 15-minute intervals; therefore, a total of 5,952 data points are available. Each data point consists of 8 features: Time, Date, Day of the week, Car Count, Bike Count, Bus Count, Truck Count, and Total. Also, each data point is categorized into four discrete classes, representing varying levels of traffic intensity: Heavy, High, Normal, and Low. For this research, the dataset is classified into two classes: Dense Traffic, which consists of Heavy and High levels of traffic intensity, and Normal Traffic, which consists of Normal and Low levels of traffic intensity.

To prepare the data for the two-stage prediction framework, extensive preprocessing and feature engineering were conducted. The raw Date and Time stamps were combined and parsed into a unified DateTime object, followed by a chronological sort to preserve temporal ordering. Essential temporal features were derived, including the Hour of the day, further cyclically encoded into Hour_sin and Hour_cos to capture its periodic nature. Vehicle counts were aggregated into a Total count, and a Total_RollMean was computed using a rolling window to identify smoothed

trends. Additionally, the Day of the week was transformed into one-hot encoded binary features. For the subsequent traffic classification stage, vehicle ratios are calculated from the predicted counts to provide context on traffic composition.

The prepared dataset is chronologically split, with the initial 80% allocated for training and validation, and the final 20% reserved as an unknown test set to evaluate model generalization. For the LSTM model, input sequences were constructed using a lookback window of 10 time steps, and all numerical features were scaled to a [0, 1] range using MinMaxScaler.

B. Model Parameter

The proposed architecture consists of two interconnected components: an LSTM network that serves as a temporal feature extractor, followed by a LightGBM classifier as a final label predictive model. This hybrid architecture is selected to combine the temporal modeling strengths of LSTM with the robust decision boundaries learned by tree-based ensemble methods.

TABLE I. LSTM HYPERPARAMETER TUNING

Hyperparameter	Range Explored	Selected Value
LSTM Units	{64, 128, 256}	128
LSTM Activation	{ReLU, tanh}	ReLU
LSTM Dropout Rate	{0.2, 0.3, 0.5}	0.3
Dense Layer Activation	{ReLU, tanh}	ReLU
Learning Rate (AdamW)	{1e-4, 1e-3, 5e-4}	1e-4

TABLE II. LIGHTGBM HYPERPARAMETER TUNING

HYPERPARAMETER	RANGE EXPLORED	SELECTED VALUE
NUMBER OF LEAVES	{31, 64}	31
MAX DEPTH	{5, 8, 10}	8
LEARNING RATE	{0.01, 0.05, 0.1}	0.05
NUMBER OF ESTIMATORS	{100, 200, 300}	200
L1 REGULARIZATION	{0.1, 0.5}	0.1
L2 REGULARIZATION	{0.1, 0.5}	0.1
FEATURE FRACTION	{0.7, 0.9}	0.9
BAGGING FRACTION	{0.7, 0.9}	0.9

The input to the LSTM is a three-dimensional tensor of shape (n, t, f) where n denotes the number of samples, $t = 10$ represents the time steps that are derived from the sliding window approach, and $f = 1$ is the number of features per time step. The LSTM subnetwork comprises two stacked LSTM layers. To determine the optimal architectural parameters, an empirical hyperparameter tuning process is conducted, varying layer widths, activation functions, dropout rates, batch sizes, and training epochs. Similarly, for the LightGBM classifier, a grid search was performed over typical ranges of tree depth, learning rates, number of estimators, and objective functions.

The explored hyperparameter ranges for the LSTM algorithm are presented in Table I. The final architecture was selected based on an optimal balance between generalization performance and training stability. The model employs 128 LSTM units with ReLU activation, accompanied by a dropout layer with a dropout rate of 0.3 to mitigate overfitting. The component is trained for a maximum of 100 epochs with early stopping, using a patience of 15 epochs. Additionally, this model utilizes adaptive learning rate scheduling via

ReduceLROnPlateau to further enhance convergence and prevent overfitting. Training was performed with the AdamW optimizer at a learning rate of $1e-4$ and a fixed batch size of 32.

Table II summarizes the hyperparameter tuning settings for LightGBM. The final configuration includes a maximum tree depth of 8, a learning rate of 0.05, and 200 estimators. The number of leaves is set to 31, while L1 and L2 regularization parameters are configured to 0.1 to control model complexity. LightGBM training and evaluation were carried out under a 5-fold cross-validation scheme, ensuring robust performance assessment across different data partitions.

IV. EVALUATION RESULTS

This section evaluates the performance of the proposed model. This section is initialized by verifying the LSTM algorithm's prediction against the temporal sequence, reporting the scores on evaluation metrics such as the RMSE, MAE, and R^2 for each vehicle category. This section also evaluates the prediction of the classification by the LightGBM model, reporting the classifier's score of accuracy, precision, and F1 score. Lastly, the results will be compared and discussed with the other ML algorithms. All experiments are conducted on 11th Gen Intel Core i5-1135G7 processor (2.40 GHz, 8 cores) and 8 GB of RAM.

A. LSTM Results

The LSTM is trained on a sequence of the last ten timesteps of all vehicle counts, i.e., cars, bikes, buses, and trucks. The dataset is split into an 80:20 train-validation split. Fig. 2 and Fig. 3 show the Huber loss and MAE Loss of LSTM training and validation.

The model's training performance, evaluated using Mean Absolute Error (MAE) and Huber Loss, demonstrates effective learning and strong generalization. Fig. 2 shows that both the training and validation losses of the Huber loss drop significantly within the first 15 epochs and converge to below 0.02. Additionally, Fig. 3 shows that both training and validation Mean Absolute Error decrease considerably during the initial 10 epochs, with the training MAE converging at around 0.10. The validation MAE follows a similar trend with minor fluctuations, indicating a balanced learning process with minimal overfitting. These results show that the model effectively represents key patterns in the data and maintains stability throughout training. Overall, the consistent downward trends and alignment between training and validation curves indicate that the model is well-regularized and properly tuned.

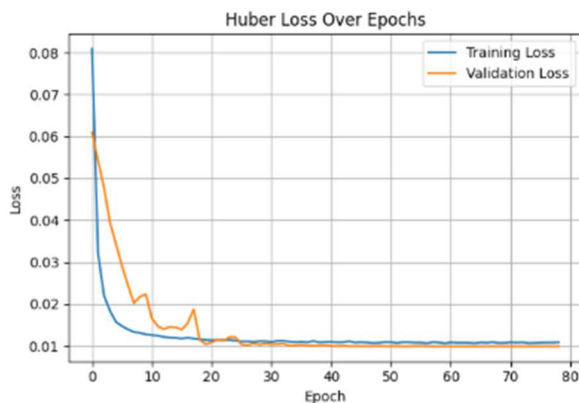


Fig. 2. Huber loss of LSTM algorithm.

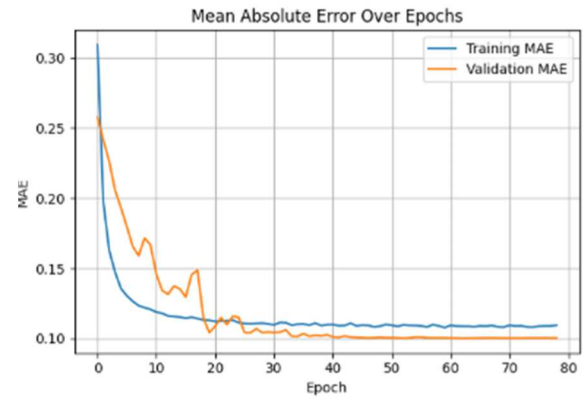


Fig. 3. Mean absolute error performance of LSTM algorithm.

To further quantify the predictive performance of the model, additional evaluation metrics, including Root Mean Squared Error (RMSE), MAE, and the coefficient of determination (R^2), are presented in Table III

TABLE III. TIME SERIES PREDICTION PERFORMANCE

Type	RMSE	MAE	R2
Car	0.12	0.09	0.79
Bike	0.10	0.07	0.71
Bus	0.15	0.11	0.71
Truck	0.17	0.13	0.57

The result shows that the MAE metric yields a relatively minimal value, with a maximum of only around 0.13 for the car type. The result indicates that the predictive model is close to the actual model. When outlier numbers are considered, the RMSE performance remains minimal. The RMSEs of the car, bike, bus, and truck are still below 0.2, indicating that the predictive model remains relatively close to the actual model, despite considering all possible outlier values. The predictive and actual time series of traffic is shown in Figure 4.

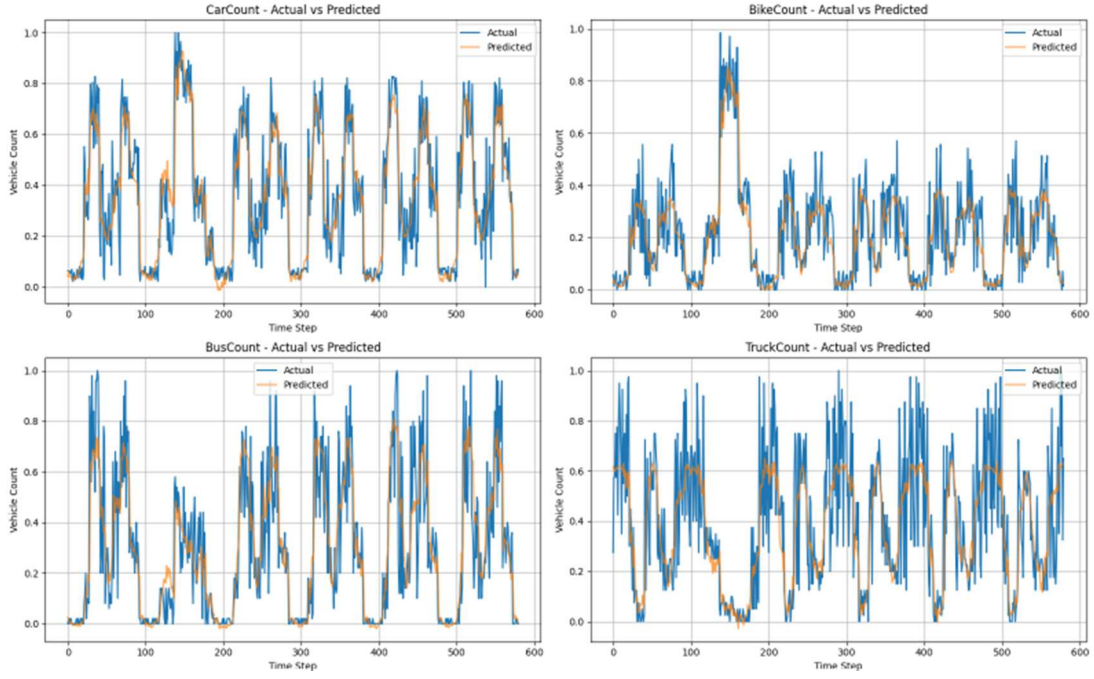


Fig. 4. The confusion matrix of lightGBM algorithm.

B. LightGBM Results

Figure 5 presents the confusion matrix of the proposed algorithm. In this framework, the LSTM algorithm produces regression outputs corresponding to predicted vehicle counts (cars, bikes, buses, and trucks). These continuous predictions are then aggregated into derived features, such as total flow and class ratios, which serve as inputs to the LightGBM classifier. At this stage, the outputs are mapped into discrete traffic states, allowing for the generation of class-level evaluation metrics.

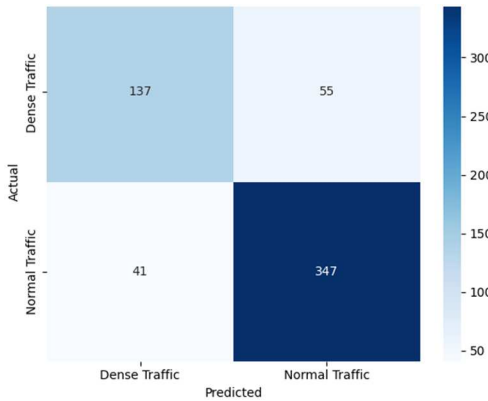


Fig. 5. The confusion matrix of LightGBM algorithm.

Table IV shows the classification reports. The proposed algorithm achieves an overall accuracy of 0.83. From each class's perspective, the evaluation shows that the precision, recall, and F1-score for Normal Traffic are higher than those for Dense Traffic. This difference can be attributed primarily to the class distribution in the dataset: Normal Traffic samples occur more frequently and cover a broader range of conditions, allowing the model to learn more stable

patterns. By contrast, dense traffic events are less frequent and often characterized by abrupt fluctuations or noisy sensor readings, e.g., sudden increases in vehicle counts, which introduce variability that is more challenging for the model to represent. Consequently, the model's ability to generalize Dense traffic cases is limited, resulting in lower precision and recall for this class. However, the similarity of results across classes indicates that the model's predictions remain reasonably balanced and do not excessively favor the majority class.

With the high accuracy of the proposed model, the proposed LSTM–LightGBM model can be further deployed on edge devices, such as roadside units or cameras, for real-time traffic monitoring. Its outputs can be fed into dashboards to support quick decisions by traffic authorities to coordinate mobility services alongside other urban data streams.

TABLE IV. CLASSIFICATION REPORTS OF THE LIGHTGBM ALGORITHM

Class	Precision	Recall	F1-Score
Dense Traffic	0.77	0.71	0.74
Normal Traffic	0.86	0.89	0.87
Macro Avg	0.82	0.80	0.81
Accuracy	0.83		

C. Comparison to Other ML Algorithms

Table V presents a comparison of various combinations of time series forecasting models and classifiers for predicting traffic conditions. For comparison, four conventional machine learning models were implemented as baselines. XGBoost is a gradient-boosting method that builds ensembles of decision trees sequentially with optimized loss functions, offering strong performance on tabular data. Random Forest constructs multiple decision trees on bootstrapped samples and aggregates their outputs, reducing variance and improving robustness. K-Nearest Neighbors (KNN) is a non-parametric method that classifies a sample based on the majority label among its closest neighbors in feature space. Logistic

Regression serves as a linear baseline, estimating class probabilities through a logistic function and providing interpretability in terms of feature coefficients.

TABLE V. COMPARISON TO CONVENTIONAL ALGORITHMS

Type	Time Series Model	Classifier	Accuracy
Proposed Algorithm	LSTM	LightGBM	0.834
Comparison Algorithm	LSTM	XGBoost	0.823
		RandomForest	0.817
		KNN	0.795
		Logistic Regression	0.782
	MLP Regressor	XGBoost	0.811
		Random Forest	0.799
		KNN	0.783
		Logistic Regression	0.770
	Linear Regression	XGBoost	0.790
		Random Forest	0.778
		KNN	0.765
		Logistic Regression	0.752

To validate the performance, this paper utilizes two other time-series prediction approaches: MLP Regressor and Linear Regression. Each time series predictive algorithm is paired with four classifiers: XGBoost, Random Forest, KNN, and Logistic Regression. Among the baseline models, the LSTM combined with XGBoost achieves an accuracy of 0.823, which is comparable to that of our proposed model. This evaluation result shows that while both boosting methods are powerful, LightGBM's optimization and leaf-wise growth strategy give it a slight edge in this context.

On the other side, MLP-based combinations achieve the highest accuracy, reaching 0.811. However, performance dropped when paired with simpler classifiers, such as logistic regression. In addition, linear regression results in the lowest accuracy across all classifier combinations, with its best accuracy reaching only 0.790, confirming its limitations in modeling complex and nonlinear time series patterns.

Overall, these findings demonstrate that the combination of a deep temporal modelling LSTM with a robust tree-based classifier, LightGBM, provides a strong pipeline for traffic classification tasks. The consistent performance shows the established architecture real-world deployment in intelligent transportation systems or urban mobility prediction platforms.

V. CONCLUSIONS

This study presents a hybrid pipeline that combines an LSTM for time series prediction with LightGBM for traffic classification. The LSTM model accurately captures temporal dynamics across various vehicle types, as supported by low error metrics, including MAE and RMSE, which are below 0.2. LightGBM further enhances the pipeline by achieving an overall classification accuracy of 0.83, outperforming other models, such as XGBoost, Random Forest, and Logistic Regression. The results show that combining deep learning for sequence modeling with a gradient boosting classifier creates a balanced and efficient solution for traffic analysis. While the current setup proves reliable performance, future research can

explore other improvements, such as integrating external features, e.g., weather or road conditions. Additionally, a low-computational optimization algorithm suitable for practical implementation can be a research direction.

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